

QA Retrieval Baseline & Failure Analysis

Research & Solution Development Intern (Take-Home Assignment)

Setup

I used a subset of SQuAD's validation set (1,500 question-context pairs, deduplicated down to 195 unique paragraphs) as a corpus to search over. I chose a subset for iteration speed, but had to fix an issue first: SQuAD's validation split is ordered by source article, so a naive small slice was almost entirely repeat questions on ~15 paragraphs. I pulled a larger slice and deduplicated contexts to get a corpus size that actually resembles a real search problem.

Baseline (Part A)

I built a TF-IDF retriever (word-overlap + cosine similarity), deliberately simple, since the goal of a baseline is correctness and reproducibility, not sophistication.

Metric	Recall@1	Recall@5	Recall@10
Score	0.615	0.871	0.921

The gap between Recall@1 and Recall@10 was the first interesting signal, it suggested the retriever often finds the right paragraph but doesn't rank it first, which pointed me toward investigating *when* it ranks things wrong rather than just how often.

Hypothesis & Finding (Part B)

I suspected failures weren't random but concentrated on questions with low word overlap with their source paragraph, since TF-IDF can only match surface words. I measured, per question, what fraction of its words also appear in the correct paragraph, then bucketed questions by that score.

Overlap bucket	n	Recall@5
Low (<0.3)	143	0.427
Medium (0.3-0.6)	705	0.864
High (>0.6)	652	0.975

A clear, monotonic trend more than a 2x accuracy gap between the extremes, affecting roughly 1 in 10 questions. But inspecting actual failing examples showed the mechanism wasn't quite what I'd assumed: most weren't paraphrase cases (different words, same meaning). They were cases where *multiple paragraphs shared the same proper nouns and topic* (e.g. several paragraphs about "the Panthers," each covering a different specific fact season record vs. defensive stats vs. play-by-play). TF-IDF can't disambiguate these because the overlapping words are identical across candidates.

Comparison Against a Semantic Retriever

To test whether this was really a lexical vs meaning problem, I built a second retriever using sentence embeddings (all-MiniLM-L6-v2) and re-ran the same bucketed evaluation:

Bucket	TF-IDF	Embeddings
Low overlap	0.427	0.755
Medium overlap	0.864	0.881
High overlap	0.975	0.923

Embeddings recovered a large share of the low-overlap failures (+32.8 points), supporting the theory that semantic similarity can disambiguate paragraphs sharing vocabulary but differing in meaning. More surprising: TF-IDF actually beat embeddings on high-overlap questions (0.975 vs. 0.923). My read is that when a question shares specific, rare terms (names, numbers, exact phrases) with its paragraph, literal matching is a stronger, more precise signal than semantic similarity, the embedding model's generalization can blur distinctions that exact matching gets right for free.

Why This Matters

There's no single "best" retriever here the right choice depends on the query. A production retrieval system likely shouldn't commit to one method globally, but could route between lexical and semantic retrieval using a signal like lexical overlap itself.

Next Step

With more time, I'd build a simple hybrid retriever combining TF-IDF and embedding similarity scores, or routing between them using the overlap score as a switch and test whether it beats both individual methods across all three buckets, not just on average.